

LESSON 4.7

COMPARING A QUANTITY AND ITS OPPOSITE IN REAL-WORLD CONTEXT



LESSON INTRODUCTION

MA.6.NSO.1.1 Extend previous understanding of numbers to define rational numbers. Plot, order, and compare rational numbers.

MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite direction using rational numbers. Compare them on a number line and explain the meaning of zero within its context

LEARNING TARGETS

- I can compare a quantity and its opposite on a horizontal or vertical line and explain the meaning of zero within the context.
- I can use the structure of a horizontal and vertical number line to plot rational numbers from a real-world context.

BENCHMARK COHERENCE

WORKING TOWARD

MA.7.NSO.2.1 Solve mathematical problems using multistep order of operations with rational numbers, including grouping symbols, whole-number exponents, and absolute value.

ESSENTIAL QUESTIONS

- How do you plot opposites on vertical or horizontal number lines?
- What can zero represent in various real-world situations?
- How does distance from zero compare for a rational number and its opposite?

LESSON NARRATIVE

In this lesson, students determine the opposites of rational numbers from a context and represent them on vertical and horizontal number lines. Students will also explain the meaning of zero and opposites within various contexts.

COMPONENT SUMMARY

4.7.1 WARM-UP (~5 MIN.)

Write and order integers that represent elevations from images

4.7.3 TRY IT (10-15 MIN.)

Plot and label rational numbers and their opposites on vertical and horizontal number lines

4.7.5 YOUR TURN! (5 MIN.)

Practice interpreting and plotting rational numbers, their opposites, and the meaning of zero in context

4.7.2 COLLABORATION (10 MIN.)

Identify and plot rational numbers and their opposites

4.7.4 BRING IT TOGETHER (5 MIN.)

Describe the relationship between rational numbers and their opposites when plotted on number lines

4.7.6 WRAP-UP (~5 MIN.)

Demonstrate understanding of the lesson learning targets

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Opposite
- Rational number

MATERIALS

- (Optional) Large paper
- (Optional) Markers

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not know the definitions of “withdrawal” and “deposit” or make sense of a deposit as losing money, as students may say, “The money would be leaving your hands so that is a loss.”

THINGS TO CONSIDER

- Check for understanding of the meaning of zero for each real-world example. For example, in 4.7.3 Try It, be sure students interpret zero as no change in account balance (not a balance of zero).

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

In 4.7.2 Collaboration and 4.7.4 Bring It Together, students make connections and comparisons among rational numbers, their opposites, and zero. Facilitate discussion about the meaning of zero in each context, as well as the comparisons and connections students make.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

Students represent rational number opposites several ways, including on number lines (horizontal and vertical), in tables (using numbers and words), and written descriptions in the 4.7.6 Wrap-Up and 4.7.3 Try It.

DIFFERENTIATE INSTRUCTION

Consider the following strategies for differentiating instruction in this lesson:

- In 4.7.4 Bring It Together, students are prompted to provide a written description explaining the relationship between the opposites. Consider having students also create a real-world example of opposites plotted on a number line to demonstrate their understanding.

4.7.1 WARM-UP: ORDERING ELEVATIONS

Component Overview: Students will analyze five pictures to determine the appropriate positive or negative integer that best represents the location. Students will then order the integers from least to greatest.

QUESTIONING

Consider challenging students by prompting them to research and provide location(s) that would have opposite elevations of the examples in the pictures in questions 1a-d. Students can also discuss if they can find opposite elevation examples with the same major regions of the U.S..



4.7.1 Warm-Up: Ordering Elevations

Four images from locations across the United States are shown.

1. Write an integer that represents each elevation.

a.  Integer: 345	b.  Integer: -100
c.  Integer: 5046	d.  Integer: -178

2. Order the letter of the pictures from lowest elevation to highest.

 d , b , a , c



4.7.2 Collaboration: A Quantity and Its Opposite

- 1.
- a. Plot the number 8 and its opposite on the number line.
- b. What do you notice about the distance of 8 and its opposite from zero?

Answers will vary. 8 and its opposite, -8, are the same distance from zero in opposite directions.

2. Complete the table for each situation given.

	78.3 Feet Above Sea Level	32°C Below Zero	Withdrawal of \$27.68
Rational Number	78.3	-32	-27.68
Opposite	-78.3	32	27.68
Meaning of Zero	Sea Level	Freezing Point 0°C	No change in account balance.

3. Omar is at the bookstore on Florida Avenue buying a book about training dogs. He also plans to visit Nolan's Pet Shop and Rover's Pet Shop to compare prices for a new leash for his dog. All three stores are on the same street. Nolan's Pet Shop is 7 blocks west of the bookstore. Rover's Pet Shop is 7 blocks east of the bookstore.
- a. Model the locations of the three stores in relation to each other on the number line.



- b. What does zero represent in this situation?
- c. Which pet store is farther away from the bookstore? Explain how you know.

Answers will vary. Both pet shops are the same distance from the Bookstore. Both locations are 7 blocks away from the bookstore.

4.7.2 COLLABORATION: A QUANTITY AND ITS OPPOSITE

Component Overview: Students collaboratively determine rational numbers, opposites, and the meanings of zero to represent real-world situations. Students also plot zero and the opposite values on number lines.

TEACHER MOVES

Compare and Connect

Students make connections and comparisons among rational numbers, their opposites, and zero. Facilitate discussion about the meaning of zero in each context, as well as the comparisons and connections students make.

ENRICHMENT

- How does the distance from zero compare for a rational number and its opposite?
- What does zero represent in this context?

4.7.3 TRY IT: A QUANTITY AND ITS OPPOSITE

Component Overview: Students plot and label rational numbers and their opposites on vertical and horizontal number lines. Students interpret the meaning of 0 and opposite values in context.

TEACHER MOVES

Facilitate conversations about the fact that zero represents the change in account balance, not the amount in the account. Conversations should also include the meaning of a deposit and withdrawal.

ENRICHMENT

- Is north up, down, left, or right? What about south?
- What does zero represent in this situation?
- How does distance from zero compare for a rational number and its opposite?

TEACHER MOVES

Consider having students write a story that includes all phrases and values. Creativity will be key in writing a cohesive narrative that incorporates all elements.

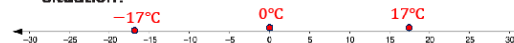


4.7.3 Try It: A Quantity and Its Opposite

- On the number line, plot and label a deposit of \$64.87 and a withdrawal for the same amount from a bank account. What does 0 represent in this situation?
Answers will vary. Zero represents no change in the account balance.



- On the number line, locate and label 17°C below zero and 17°C above zero. What does 0 represent in this situation?



Zero represents a temperature of 0°C , freezing point.

- Carlos is at the ice cream shop. His house is 9.5 blocks north of the shop. The park is 9.5 blocks south of the ice cream shop.
 - Plot and label all three places on the number line.
 - When at the ice cream shop, is Carlos closer to the park or his house?

Answers will vary. The ice cream shop is equal distance to both locations. Ice cream shop represents zero.



- Write the rational number that represents the description, the rational number's opposite, and the meaning of the opposite in context of the description.

Description	Rational Number	Opposite	Meaning of Opposite
An atom's positive charge of 7	7	-7	An atom's negative charge of 7
A deposit of \$35.84	35.84	-35.84	A withdrawal of \$35.84
3,540 feet below sea level	-3540	3540	3540 feet above sea level
A rise of 41.6°C	41.6	41.6	A fall of 41.6°C
A loss of 13.3 pounds	-13.3	13.3	A gain of 13.3 pounds

- With a partner, choose one description and describe what 0 could represent in that situation.

*Answers will vary.
An atom with no charge.
No change in an account balance.
A location at sea level.
No change in the temperature.
No change in weight.*

**4.7.4 Bring It Together: A Quantity and Its Opposite**

Work with a partner to write answers to the following questions.

1. What is the relationship between any number and its opposite when plotted on a number line?
Answers will vary. A number and its opposite are the same distance from zero on a number line, but on opposite sides of 0.
2. How can you use this relationship to locate the opposite of a given number on the number line?
Answers will vary. To find the opposite of a given number on a number line, find the same distance, away from zero, in the opposite direction.

4.7.4 BRING IT TOGETHER: A QUANTITY AND ITS OPPOSITE

Component Overview: Students work with a partner to describe the relationship between rational numbers and their opposites when plotted on number lines. Select a few groups to present their responses during the whole-class debrief discussion.

TEACHER MOVES

Compare and Connect

Students make connections and comparisons among rational numbers, their opposites, and zero. Facilitate discussion about the meaning of zero in each context as well as the comparisons and connections students make.

4.7.5 YOUR TURN!

Component Overview: Students practice determining and plotting rational numbers, their opposites, and the meaning of zero in context.

ENRICHMENT

- Is the recommended temperature actually zero?
- What does that make you think about when engaging with absolute value around numbers that are not zero?
- Can a number line still be used?



4.7.5 Your Turn!

1. Janney notices that her room's temperature is 4°F below the recommended temperature, and Ivan's bedroom temperature is 4°F above the recommended temperature. Plot the values on the number line and explain what zero represents in this situation.

Answers will vary. Zero represents the recommended temperature.



2. Finn wants to visit his friend Simon who lives 22.7 blocks west. He also wants to visit his grandparents who live 22.7 blocks east.
 - a. Plot and label the locations of Simon and Finn's grandparents on the number line.



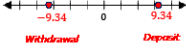
- b. What does zero represent in this situation?
Answers will vary. Zero represents Finn's house.
- c. Does Finn live closer to his grandparents or Simon?
Answers will vary. Finn's house is equal distance to both locations.



4.7.6 Wrap-Up: Ticket Out the Door

Complete the following in the table.

- Write the rational number that represents the situation.
- Write the opposite of the rational number.
- Represent the rational number and its opposite on the number line with appropriate labels.
- Explain the meaning of zero in the situation.

	A Deposit of \$9.34	17°C Below Zero
Rational Number	9.34	-17
Opposite	-9.34	17
Number Line		
Meaning of Zero	There was no change in the account balance.	Freezing point. 0°C

4.7.6 WRAP-UP: TICKET OUT THE DOOR

Component Overview: Students demonstrate understanding of the lesson learning targets. Use data from this formative assessment to determine which students need remediation before the next lesson.

TEACHER MOVES

Challenge students to create a context other than what was already used in the lesson and provided here. If students are stuck, prompt students to determine 0 as a certain location in their neighborhood and choosing a new location for the value of 9.34. For example, if their house is 0, then 9.34 could be the distance to the park, while -9.34 could be the distance to their local store (with units).